Approval-Based Elections in the 1D Euclidean Domain

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Abstract

We study approval-based elections, that is, elections where each voter approves a subset of candidates. Our focus is on preference restrictions, in particular on a recently introduced model: the voter/candidate range, defined in the work of Godziszewski et al. (2021). We explore multiple areas: First, we create a domain detection algorithm based on a reduction to integer linear programming, and compare it with existing solutions. Next, we use the said algorithm in an experiment, in order to establish the probability with which a random election falls within a certain domain. At last, we provide a polynomial-time algorithm for the approval-voting constructive control problem, for elections in the one-dimensional voter/candidate range domain.

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Chapter 1

Introduction

Elections are a ubiquitous element of our society in one form or another. For example, societies need to choose members of parliaments, judges have to pick winners of sport competitions, or citizens need to elect mayors of cities. However, elections are not only applicable to politics or competitions. In more day-to-day examples of elections, teams have to schedule meetings times, video streaming platforms need to provide lists of recommended movies to the users, or a company needs to allocate a yearly budget. Each of these problems may require a different type of an election and a different strategy of selecting the winner (or winners). The differences come from the underlying properties and objectives of the problem. For example, in presidential elections we want to choose the candidate supported by the majority of the voters. It is not necessarily the case for the parliamentary elections, where the goal is to select members of the parliament in such a way that the opinions of the voters are represented proportionally. Another example would be the problem of scheduling a meeting, where we would try to pick the time that, maximizes the number of participants who can attend, even though, the selected time might not be optimal for anyone (e.g., early morning). In sport competitions, especially in the seasonal ones, i.e., the ones consisting of series of events, we need to come up with a strategy of calculating overall ranking at the end of the season, based on the aggregated results from the particular events. For example, in FIS Alpine Ski World Cup, or in Formula 1 World Championship, the overall winners achieve very high scores consistently throughout the season, but might not have necessarily achieved the highest scores (1st place) in the majority of the individual events.

Another key distinction between the considered examples, is whether a given election has just one winner, or requires selecting multiple candidates. More formally, in computational social choice these election types are called *single-winner* or *multiwinner* elections. Multi-winner (committee) elections are even more widespread than the single-winner ones, and in most cases they have different properties and require different voting rules. For an overview

of various committee election types and their related properties, see the works of Faliszewski et al. (2017) and Elkind et al. (2017). A committee election is the process of choosing a fixed-size subset of candidates (i.e., the committee), based on the voters' preferences. Voters' ballots (preferences) can be represented in various different ways. In this work we focus on approval elections, that is, on elections where each voter provides a subset of candidates that he or she approves of. Another notable type of elections is the ordinal model, where the voters order the candidates from the most to the least preferred ones. For an extensive survey of the approval model and current state of the art regarding committee elections, we point the reader to a survey by Lackner and Skowron (2020).

Computational social choice theory has a wide variety of problems it tries to solve. Above-mentioned examples consider a winner determination problem. It is one of the core problems, however not the only one. Other ones include, e.g., control problems (see, e.g., the works of Bartholdi et al. (1992); Hemaspaandra et al. (2009)), which refer to certain ways of manipulating elections. In short, we try to make a given candidate win (or lose) by changing the structure of an election, that is, by adding or deleting voters or candidates. Other interesting problems, that we are not going to explore in this work, include manipulation (see, e.g., the works of Bartholdi et al. (1989); Faliszewski and Procaccia (2010), and the survey of Conitzer et al. (2016)), bribery (see, e.g., the works of Faliszewski et al. (2006), and the survey of Faliszewski et al. (2016)), justified representation (Aziz and Walsh (2014)) and many others (see the handbook edited by Brandt et al. (2016)).

1.1 Preference Restrictions

Many of the above-mentioned social choice problems are known to be computationally hard. We can deal with this complexity using various methods, such as designing approximation algorithms (random or deterministic) (see the work of Vazirani (2002)) or parametrized algorithms (see the book by Cygan et al. (2015)). An alternative approach is to introduce restrictions on the preference domains. In other words, we restrict the structure of the voters' ballots. There is a successful and active line of research in this area. It begins with the notions of *single-peaked (SP)* (Black (1958); Arrow (1951)) and *single-crossing (SC)* (Mirrlees (1971); Roberts (1977)) elections. Much more recently, Barberà and Moreno (2011) introduced the concept of *top-monotonicity*, a generalization of the SP and SC restrictions. These models provided a basis for numerous and promising results in circumventing the computational complexity in a variety of problems. (Walsh (2007); Faliszewski et al. (2009); Skowron et al. (2015); Brandt et al. (2015); Elkind et al. (2016)). Naturally, many other restricted domains were also studied.

In the context of this work, we emphasize a survey by Elkind and Lackner (2015). Unlike most of the above-mentioned examples, which considered the ordinal model of preferences, the survey explored well-known preference restrictions in the setting of approval voting, and introduced several new ones. This includes models such as *Voter Interval (VI)* and *Candidate Interval (CI)*, both based on the *Dichotomous Euclidean (DE)* restriction. Euclidean models are very intuitive, a simplified explanation of such models is that each candidate and each voter is assigned an ideal point in some Euclidean space $\mathbb{R}^d$, this point represents their opinion on some d issues. Furthermore, each voter has an additional radius parameter. The points are assigned in such a way that voters approve the candidates that are within their radius, and prefer the closer ones. Godziszewski et al. (2021) extended this model, making it more versatile, by using a radius in addition to a point, to represent both the candidates and voters. They named the new model a *Voter/Candidate Range (VCR)* model. We note that the VI and CI models are special cases of the one dimensional VCR model. However, Godziszewski et al. (2021) use different terminology, and their names for the VI and CI are the one dimensional *Candidate Range* (1D-CR) and one dimensional *Voter Range* (1D-VR) models, respectively.

1.2 Goals and Contributions

The goal of this work is to study the one-dimensional Voter/Candidate Range model (1D-VCR), and to compare it with the 1D-CR and 1D-VR models. Our main contributions are as follows:

1. In Chapter 3, we provide detection algorithm for the 1D-VCR property, based on a reduction to integer linear programming (ILP). Additionally, we apply it to detecting 1D-VR and 1D-CR profiles and, later, we do a performance comparison with a solution based on a 2-SAT reduction due to Fitzsimmons and Lackner (2019).

2. In Chapter 4, we analyze the 1D-VCR domain and compare it to other well known models. We use multiple statistical cultures to generate random elections in the 1D-VCR model, then we check how many of them do not satisfy 1D-VR or 1D-CR properties. Furthermore, we inspect how different are the 1D-VCR elections from the 1D-VR and 1D-CR ones.

3. In Chapter 5, we propose a polynomial time algorithm for a certain control problem under the 1D-VCR model, expanding on the solutions for the 1D-CR and 1D-VR models, due to Magiera and Faliszewski (2014) and Faliszewski et al. (2009), respectively.

We do not discuss the related work right now but, instead we do it throughout this thesis, as the discussion will greatly benefit from the added context and definitions introduced in the subsequent chapters.

Chapter 2

Preliminaries

First, in Section 2.1, we introduce basic definitions and core concepts, together with simple examples. Later, in Section 2.2, we discuss preference restrictions, and how can we leverage them to design better algorithms, for many of the computational social choice problems. Finally, in Section 2.3, we explain some computational methods (tricks) we have used in this work, that we believe are not necessarily straightforward.

2.1 Basic Definitions

For an integer t, by $[t]$ we mean the set $\{1,...,t\}$. To describe a matrix $\mathbf{A}$ with r rows and c columns, we use $\mathbf{A}_{r\times c}$. By $\mathbf{A}[i,j]$ we refer to the entry in i-th row and in j-th column of a matrix $\mathbf{A}$.

Approval Elections

An approval-based election (in short, an *election*, or interchangeably a *profile*) is a pair $E = (C,V)$, where $C = \{c_1,...,c_M\}$ is a set of candidates and $V = \{v_1,...,v_N\}$ is a set of voters. We use M and N for the number of candidates and voters respectively. For each voter $v \in V$, his or hers preferences are represented as a set of approved candidates $A(v)$, we call it an *approval set*. Furthermore, for each candidate $c \in C$, we define $V(c) = \{v \in V \mid c \in A(v)\}$, a set of voters who approve candidate c.

In practice, we often use an additional, simplified notation of representing an election. We redefine A as a binary matrix (a matrix, where each entry is in $\{0,1\}$), with N rows and M columns. For $i \in [N]$ and $j \in [M]$, entry $A[i,j]$ is 1 if voter v_i approves of candidate c_j, and 0 otherwise. In other words, each row of the matrix is a vector representation of a given voter's

approval set. We refer to it as an *approval matrix* or a *preference matrix*. It will always be clear from the context which notation we are using.

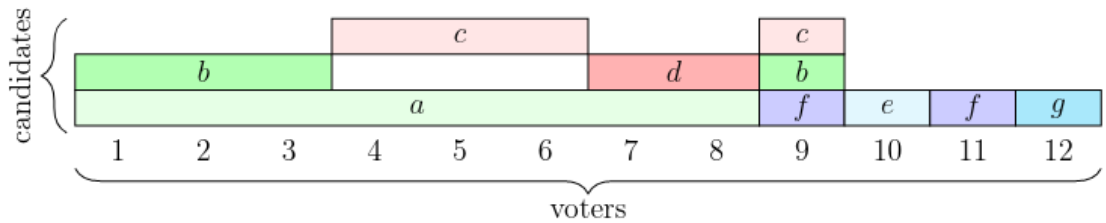


Figure 2.1 Example of an approval election. A candidate is approved by a voter if his or hers block is above the voter's point. Temporary image from Lackner and Skowron (2020)

Example 1. *(WIP: Opis wyborów z figury) Following the example from Figure 2.1 ...*

Multiwinner Committee Selection Rules

Voting rules are algorithms used to, select a committee (winners) — fixed size subset of the candidates, based on the voters preferences, for a given election. Each rule has different properties and must be selected according to the goal of the election. As this is not the main focus of this work we point to Faliszewski et al. (2017), for a detailed overview of multiwinner voting rules.

Nonetheless, we introduce the most basic voting rule for the approval model, simply called *Approval Voting (AV)*, introduced by Brams and Fishburn (1983).

Definition 2.1.1 (Multi-Winner Approval Voting (AV))**.** *For a given election $E = (C, V)$ and a size of committee k, AV rule selects the k candidates with the highest approval scores. AV-score of a given candidate c, is the number of voters who approve of c, and formally, is defined as $score_{AV}(E, c) = |\{v \in V : c \in A(v)\}|$.*

Example 2. *(WIP: Opis wyborów z poprzedniego przykładu + kto wygra) Continuing with the election from Example 1*

2.2 Preference Restrictions

As a reminder from Section 1.1, many computational social choice problems such as winner determination, or control problems are known to be computationally hard. Preference restrictions, i.e., restrictions on the structure of voters' ballots, are a very effective solution to circumvent the computational complexity of these problems.

2.2.1 Euclidean Preference Restrictions

Euclidean preference restrictions, in the approval voting model, are foundation of this work. All the problems we solve in the subsequent chapters are structured around this concept. Unless stated otherwise, throughout this work, we consider one-dimensional variants of the Euclidean models.

Let us introduce restrictions that we explore in this work. Starting with the most basic one, that is a predecessor to the other ones, in this family of restrictions.

Restriction 1 (Dichotomous Euclidean (DE))**.** *[Elkind and Lackner (2015)]*

Given an election $E = (C,V)$, we say that E satisfies DE if there is a mapping ρ of voters and candidates into the real line such that for every voter $v \in V$ there exists a radius r_v, and it holds that $v = \{c : |\rho(v) - \rho(c)| \leq r_v\}$. (WIP: Czy w ogóle definiować DE?, czy jest potrzebna cytacja?, ta definicja nie jest specjalnie spójna z pozostałymi oznaczeniami / nie oczywista)

The DE preference restriction is somewhat limited, it permits for only one agent type, i.e., candidate or voter, to have a radius parameter, while the other one is represented just by his or hers point. We consider the DE model in one of the two variants:

1. The voter-range model, where voters have the radius parameter. In other words, if a voter approves a candidate then he or she also approves all the closer ones.

2. The candidate-range model, where candidates have the radius parameter. We say that if a candidate is approved by a voter, then he or she is also approved by all the voters closer to him or her.

Below we provide a formal definition of these two models, as well as their generalization which allows all agents to have a radius parameter.

Restriction 2 (Voter/Candidate Range (VCR))**.** *[Godziszewski et al. (2021)]*

Given an election $E = (C,V)$, we say that the voters have one-dimensional Euclidean preferences if for each agent $a \in C \cup V$ (i.e., for each candidate and each voter) there exists a point $x_a \in \mathbb{R}$ and nonnegative real value $r_a \in \mathbb{R}$ such that:

$$c \in A(v) \iff |x_c - x_v| \leq r_c + r_v$$

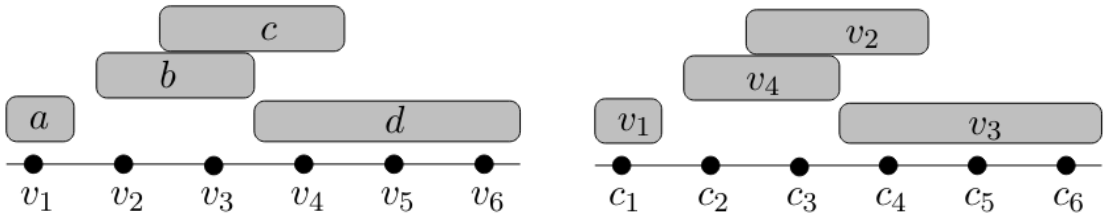
We can think of the VCR restriction as for an agent $a \in C \cup V$ the point x_a describes a's ideal position in $\mathbb{R}$, representing his or hers opinion. For a candidate $c \in C$, radius r_c can be seen as c's charisma, or in other words their ability to reach a number of different voters

(e.g., a centrist candidate possibly would have a larger reach, than an extremist one). For a voter $v \in V$, radius r_v corresponds to v's willingness to compromise, that is, how different (faraway) opinion they would still accept from their ideal one. Moreover, there are two special cases of the VCR model.

Restriction 3 (Candidate Range (CR))**.** *We say that a VCR election $E = (C,V)$, is CR if for each voter $v \in V$, their radius r_v is equal to zero.*

Restriction 4 (Voter Range (VR))**.** *We say that a VCR election $E = (C,V)$, is VR if for each candidate $c \in C$, their radius r_c is equal to zero.*

The CR and the VR models are equivalent to the voter interval (VI) and the candidate interval (CI) ones, both introduced by Elkind and Lackner (2015). Godziszewski et al. (2021) simply use a different terminology.



(a) CR election (b) VR election

Figure 2.2 Example of elections with the CR and the VR preference restrictions. Temporary Image from Elkind and Lackner (2015)

The VCR domain is larger than the CR and the VR domains combined. Below in Example 3, we show an election that is VCR, but neither CR nor VR. We explore this topic further in Chapter 4.

Example 3. *Given an election E with the candidates $C = \{a,b,c,d\}$ and the voter's approval sets $\{a,b\}$, $\{b,c\}$, $\{b,d\}$, and $\{a,b,c,d\}$.*

To see why, ... (WIP: Podać punkty i promienie (VCR), następnie opisać dlaczego nie jest CR ani VR ... Obrazek?)

2.3 Integer Linear Programming

Integer linear programming (ILP) is a common approach to solve NP-hard problems, furthermore it can be used successfully for other problems as well. To solve a problem by

reduction to the linear program, first we need to define the given problem as a system of linear inequalities and an objective function. Formally, for $j \in [n]$, a single inequality consists of a set of decision variables x_j, a set of parameters a_j and a constraint b, together giving a formula: $a_1x_1 + a_2x_2 + ... + a_nx_n \leq b$ However, in practice there are multiple constraint inequalities, and we use more general notation to represent an i-th inequality:

$$\sum_{j=1}^{n} a_{ij}x_j \leq b_i$$

Besides constraints, we define an objective function with parameters c_j:

$$\sum_{j=1}^{n} c_jx_j$$

Now, having defined inequalities and the objective, the goal is to find a vector of values for the x_j, such that it maximizes (or minimizes) the value of the objective function, while keeping all the inequalities true. In general, all the variables of a linear program can be fractional, and such program can be solved in polynomial time. However, there is a special case called *(mixed) integer linear program*, that is, when we require some variables to be integers. Solving integer linear programs is NP-hard. Despite this, there are available state-of-the-art linear program solvers, that solve ILPs effectively. In this work we use one of them, the Gurobi solver. For the complete theory of (integer) linear programming, we point to the book by Wolsey and Nemhauser (2014).

Not every problem can be directly defined as an integer linear program. However, we can apply basic techniques and tricks to make the problem suitable. For example, this is the case for our solution to the VCR profile detection problem described in Chapter 3. Below we present these methods:

Big-M Method. This technique is used to overcome a variety of problems, such as representing absolute or minimum/maximum values as linear inequalities. Another examples would be defining a logical conditions (And/Or) between the inequalities or using indicators (i.e., if-then logic). These problems are not linear in their nature. However, by taking certain steps we can linearize them, and then use them in a linear program.

Having a set of m linear inequalities with n variables represented as:

$$\sum_{j=1}^{n} a_{ij}x_j \leq b_i \qquad i \in [m], j \in [n]$$

The fundamental concept of the BigM method is to introduce to the inequalities, an additional large constant coefficient M (hence Big), which is greater than the maximum value of any variable x_j. Based on how we do it decides which of the aforementioned problems we solve. Below in Example 4, we showcase the BigM technique to represent *OR* condition between inequalities.

Example 4. *We have two inequalities and want to represent a logical OR condition between them.*

$$\sum_{j=1}^{n} a_{1j}x_j \leq b_1 \quad or \quad \sum_{j=1}^{n} a_{2j}x_j \leq b_2.$$

To do so, we introduce a large constant M and two binary variables $y_1, y_2 \in \{0,1\}$. *First we transform the inequalities:*

$$\sum_{j=1}^{n} a_{1j}x_j \leq b_1 + M(1-y_1)$$

$$\sum_{j=1}^{n} a_{2j}x_j \leq b_2 + M(1-y_2)$$

Then, we add one more constraint, binding the inequalities together:

$$y_1 + y_2 \geq 1$$

This constraint enforces that at least one of the inequalities will be true.

The BigM method is a valuable and widely used tool, however it has some downsides, mostly related to the numerical issues. That is why, most ILP solvers provide additional built-in mechanisms for applying it. For further reading, we point to the reference manual of Gurobi Optimization (2021). We use one of those mechanisms for our linear program in Chapter 3, namely an *indicator constraint*. It allows one to create constraints representing if-then logic. Formally, we have to introduce a binary variable $z \in \{0,1\}$, and then we can define an indicator constraint in the form of:

$$z = f \longrightarrow \sum_{j=1}^{n} a_{ij}x_j \leq b_i$$

The constraint states that if variable z is equal to $f \in \{0,1\}$, then the linear constraint should hold. On the other hand, if $z = 1 - f$, then the linear consist may be violated.

Chapter 3

Voter/Candidate Range Profile Detection

First, in Section 3.1, we discuss various approaches to preference recognition, and we present selection of related work. Followed by Section 3.2, where we describe our main contribution of this chapter. That is, an algorithm based on a reduction to ILP for deciding if a given approval profile has the VCR structure. Finally, in Section 3.3 we experimentally compare the performance of our ILP based solution with the one based on a reduction to 2-SAT, for a task of recognizing the CR and the VR preference restrictions.

3.1 Profile Structure Detection

To take advantage of the preference restrictions defined in Section 2.2 or any other one, we need to design detection (recognition) algorithms. They can be used to check whether a given election has particular structural properties. Research in this area is active and successful. Provided solutions can be divided into two categories. First one, where preference recognition algorithm is based on a certain characterization of a domain. For example, Peters and Lackner (2020) show that a profile is *single-peaked on a circle* by defining a forbidden-substructure of profiles. For the *single-peaked single-crossing* domain, Elkind et al. (2020) characterize profiles by connecting them to other smaller domain. Second category is more vague, and the solutions are somewhat conventional. For example, Doignon and Falmagne (1994) provide detection algorithm for the *1d-Euclidean* domain. Magiera and Faliszewski (2019) reduce the *top-monotonicity* recognition problem to the SAT-2CNF problem. Above-mentioned examples consider elections in the ordinal model, nonetheless, there are recognition algorithms for the approval-based elections as well. For a large group of preference restrictions among others, the CR and the VR, the problem of recognition can be reduced to detecting consecutive ones property (Definition 3.1.1) in the approval matrix of a profile (Elkind and Lackner (2015)). However, for some restrictions designing a polynomial

time domain recognition algorithm is not possible. Peters (2017) showed this to be true for the case of *d-Euclidean* ($d \geq 2$) restrictions in the ordinal election model.

Definition 3.1.1. *A binary matrix (i.e., a matrix consisting of zeros and ones) has the consecutive ones property (C1P or COP) if its columns can be permuted in such a way that for each row the ones are consecutive, i.e., the 1s form an interval.*

In this work we refer to the term C1P more broadly, using it interchangeably, for an analogous scenario where we permute the rows and look for the consecutive 1s in every column in a given matrix. Booth and Lueker (1976) provided a linear-time C1P detection algorithm based on the PQ-Tree data structure.

Proposition 1 (Elkind and Lackner (2015))**.** *There is an algorithm that given an approval election with m candidates and n voters decides if it satisfies the VR (the CR) condition in $O(mn)$ time.*

Proof. Let us represent the election as an approval matrix $A_{n\times m}$. We can see that for the VR property, permuting the columns of matrix A so that ones form an interval in each row is equivalent to permuting candidates so that the set of candidates approved by each voter forms an interval. For the CR case, we use our extended definition of the C1P (the transposed variant), where we permute the rows so that ones form an interval in each column, which is equivalent to permuting voters so that the set of voters who approve a certain candidate forms an interval. □

In this work, our focus lies on designing an algorithm for recognition of the one dimensional VCR property. That is, having an approval matrix as input we have to decide if for each agent (i.e., for each voter and each candidate) there is a point on the real line and a nonnegative real value representing the radius, so that these parameters satisfy the VCR property.

3.2 Integer Linear Program

An algorithm based on integer linear program (ILP) solving is an excellent choice for the first attempt at designing a detection algorithm. It provides an efficient and relatively simple solution. Furthermore, it can be used as a fast benchmark for validating future solutions. Below we provide an ILP for the VCR profile recognition problem, together with additional preliminary steps.

Transforming the Definition of VCR into Linear Constraints

Before formulating an ILP for detecting VCR profiles, we have to redefine the definition of VCR in order to make it appropriate for an ILP solver. As a short reminder from Section 2.2.1, for the candidate set C and voter set V, we say that the 1D-VCR property is satisfied if for each candidate $c \in C$ and each voter $v \in V$:

$$c \in A(v) \Leftrightarrow |x_c - x_v| \leq r_c + r_v,$$

where x_a and r_a represent agent's point on the real line and their radius, respectively. The definition is based on the Euclidean distance which cannot be directly represented in an ILP. Equation 3.1 shows transformation of the VCR definition into linear inequalities.

$$\begin{aligned} &|x_c - x_v| \leq r_c + r_v \iff \\ &-(r_c + r_v) \leq x_c - x_v \leq r_c + r_v \iff \\ &x_c - x_v \leq r_c + r_v \;\; \vee \;\; -(x_c - x_v) \leq r_c + r_v \end{aligned} \tag{3.1}$$

The constraint obtained from Equation 3.1 is not suitable yet for an ILP because it contains a logical *OR* condition, which again cannot be directly represented in an ILP. To transform it into linear form, we use special type of the constraints provided by the Gurobi solver, i.e., indicator constraints, which allow representing if-then logic. They are based on the so-called BigM technique, which we described in Section **??**.

Detection Algorithm

We present our final formulation of the ILP for recognising VCR elections in Figure 3.1. The program takes as input preference matrix A, and if a solution exists it outputs every agent's position x_a and radius r_a. Additionally, we use two binary variables p and q, required by the indicator constraints.

Variables :

$$x_a \in \mathbb{R} \qquad a \in C \cup V$$

$$r_a \in \mathbb{R}^{0+} \qquad a \in C \cup V$$

$$p_{c,v} \in \{0,1\} \qquad c \in C,\ v \in V$$

$$q_{c,v} \in \{0,1\} \qquad c \in C,\ v \in V$$

Constraints :

$$\text{for } c \in A(v)$$

$$x_c - x_v \leq r_c + r_v$$

$$-(x_c - x_v) \leq r_c + r_v$$

$$\text{for } c \notin A(v)$$

$$p_{c,v} = 1 \rightarrow x_c - x_v \geq r_c + r_v + 1$$

$$q_{c,v} = 1 \rightarrow -(x_c - x_v) \geq r_c + r_v + 1$$

$$p_{c,v} + q_{c,v} >= 1$$

Figure 3.1 An ILP for detecting the VCR property. Variables x_a and r_a represent agent's position and radius, and variables $p_{c,v}$ and $q_{c,v}$ are required to use indicator constraints.

3.3 Performance comparison of ILP and SAT based solutions for COP detection

In this section we propose a small extension of our ILP, so it can be used to detect the VR and the CR properties. Later, we compare it with an alternative solution, which is based on a reduction to the 2-SAT problem, due to Fitzsimmons and Lackner (2019).

Extending the VCR recognition ILP

We know that the VR and the CR properties are special cases of the VCR restriction, to be specific, they require all candidates or all voters to have radius equal to zero, respectively. The extension is straightforward, we need to introduce additional constraints that require each agents' (candidates' or voters') radius to be zero. For the VR case it is: $r_c = 0 \quad c \in C$, and for the CR: $r_v = 0 \quad v \in V$.

Comparison experiment

The 2-SAT based solution is an algorithm for the consecutive ones property detection in a matrix. From Proposition 1 we know it is equivalent to detecting the VR property in an approval matrix of an election. With that said, we expect the ILP solution to be slower than the 2-SAT one, the latter one is specialized in one task only, that is detecting the COP. On the other hand, the ILP solution is more general, as it outputs not only a VR certificate, but also a mapping of agents to points and radii. We design a simple experiment to find out how much faster is the 2-SAT solution.

We conduct the experiment in the following steps:

1. We consider three elections sizes. Having m candidates and n voters, election dimensions are: $(m,n) \in \{(15,15),(20,20),(40,40)\}$.

2. We generate 100 random VCR profiles for each election size (generation details are discussed in Section 4.1.2).

3. For both algorithms, we measure time of the model creation and optimization to detect the VR property. To minimize external factors on measurement time, we repeat this step three times, and pick the smallest result.

Results. Let us analyze the results visualized in Figure 3.2. Each subplot shows results for a given election size. Algorithms are represented on y-axis, and measured detection time on x-axis in seconds. Furthermore, each point is colored based on the detection result. 2-SAT based algorithm is significantly faster and more consistent, it is anywhere between 10 and 100 times faster. These conclusions are reflected in statistical metrics presented in Table 3.1. Additionally, we have performed one-off measurements for some larger elections (i.e., $80C \times 80V$ and $100C \times 100V$), for which detection time differences were even bigger.

Algorithm	Election Size	mean	std
ILP	15 Candidates 15 Voters	0.113	0.024
	20 Candidates 20 Voters	0.211	0.047
	40 Candidates 40 Voters	2.055	0.924
SAT	15 Candidates 15Voters	0.004	0.001
	20 Candidates 20 Voters	0.012	0.006
	40 Candidates 40 Voters	0.037	0.063

Table 3.1 Statistical metrics for detection time, grouped by algorithm type and election size.

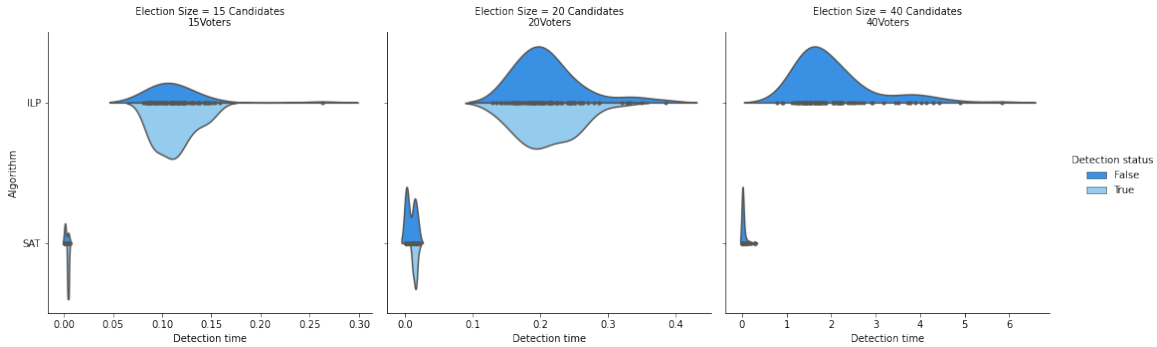


Figure 3.2 Violin Plot: VR (COP) Detection time comparison between ILP and SAT. Results are divided into 3 subplots based on the election size. Categories, that is algorithms used (i.e., ILP or SAT) are described on y-axes. On x-axes detection time is shown in seconds. Additionally, we distinguish detection outcome (i.e., positive or negative) of each measurement with different colors. Precise statistical metrics are included in Table 3.1.

3.4 WIP: Conclusions

2SAT is great, but we have not studied VCR domain enough to construct one

Chapter 4

Voter/Candidate Range Profiles

In this chapter we explore the Voter/Candidate Range domain, describe some of its properties, and compare it with the Voter Range and Candidate Range ones. We already know that the VCR domain is larger than the VR and CR ones combined (Section 2, Example 3). However, we would like to know how much bigger it actually is, and if we can significantly benefit from further development of algorithms based on the VCR model.

4.1 Domain analysis

We have performed several experiments, in each of them we check how often a profile has the TVCR restriction, compared to the other ones. For the first experiment we focus on the small size elections and only observe the overview of the properties. For the second one, we do a more thorough examination. We check larger profiles, and compare the properties in detail. In every experiment we use the detection algorithm described in Section 3.2.

For the sake of text clarity and more explicit naming, we introduce additional notations to describe some special cases of the VCR preference restriction.

Definition 4.1.1. *A profile P is **Truly VCR (TVCR)**, if P satisfies the VCR property and does not satisfy either of the VR and the CR properties.*

Definition 4.1.2. *A profile P is **Fully COP (FCOP)**, if P satisfies both the VR and the CR properties (see Definition 3.1.1, for a reminder of the COP).*

4.1.1 Experiment 1: How many profiles are TVCR or VCR?

For this experiment we performed a complete search of the domain. That is, we generated every possible profile of a given size and checked if it has the VCR property,

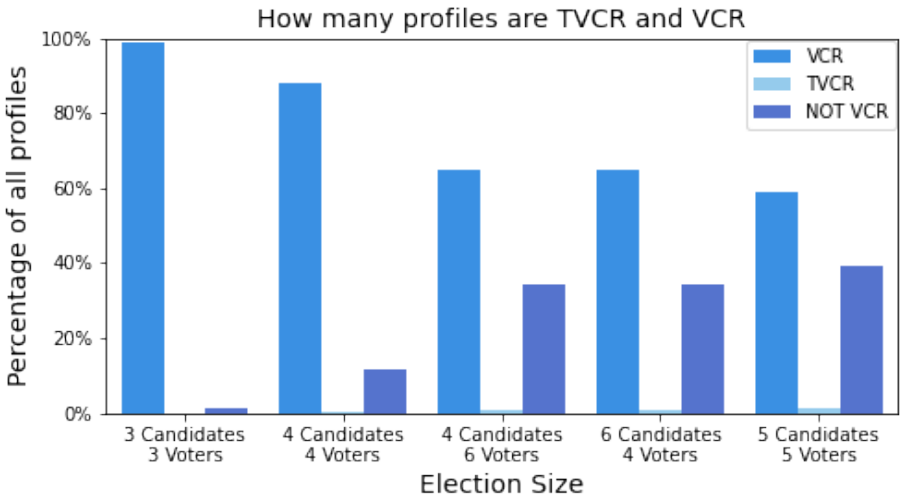


Figure 4.1 Relative number of the profiles with a given property (VCR and TVCR), out of all possible ones for a given election size, from Experiment 1

and if so we checked again if it was the TVCR variant. We focused on the small size of elections, having elections of M candidates and N voters, we considered $(M,N) \in \{(3,3),(4,4),(5,5),(4,6),(6,4)\}$.

The results are presented in Figure 4.1. They show that the VCR domain is relatively large. However, for these small election sizes, only a limited number of profiles are Truly VCR (i.e., neither VR nor CR). For example, looking at the two smallest elections, for the 3 candidates, 3 voters case, there is no TVCR profiles at all. For the 4 by 4 case, there is only 96 of them, out of 2^{16} profiles. Another observation is a decrease in a number of the VCR profiles, in favor of the not VCR ones, with the increase of the election size (e.g., for the 5 candidates, 5 voters case, only about 60% of profiles are VCR).

4.1.2 Experiment 2: How many profiles are TVCR, CR, VR or FCOP?

In this experiment, we only consider the VCR profiles. We want to find out, how their numbers are distributed across different variants of the VCR restriction. We generated a suitable number of random VCR profiles and checked how many of them are neither CR nor VR (TVCR). We grouped the results into three additional categories, besides the TVCR one, that is, VR and CR (FCOP), VR and not CR, CR and not VR.

We employ several strategies to generate a large number of profiles with distinct characteristics. We use various statistical cultures (distributions) to pick random points and radii representing the candidates and voters. Furthermore, we choose the random distribution independently for the position and radius parameters, and independently for the candidates

and voters. We describe these categories in detail below. For each category pair (in total, 20 pairs) we generate 1000 elections with 20 candidates and 20 voters. Then we repeated the experiment for the elections consisting of 40 candidates and 40 voters.

Position Parameter

We consider four ways of generating the points, representing the positions of the candidates and the voters:

1. The *Uniform Candidate Positions, Uniform Voter Positions (UCP/UVP)* model, where the points for both, the candidates and the voters are selected uniformly at random from the $[-10, 10]$ interval.

2. The *Asymmetric Gauss Candidate Positions, Asymmetric Gauss Voter Positions (GCP/GVP)* model, where we take 70% of points from a Gaussian distribution with mean value -3 and standard deviation 1.8, and the remaining 30% come from a Gaussian with mean 3 and the same standard deviation. This model is inspired by the 2 dimensional model from the work of Godziszewski et al. (2021). The goal is to simulate a society with a large majority, and a minority with the opposite opinion.

3. The *Uniform Candidate Positions, Asymmetric Gauss Voter Positions (UCP/GVP)* model, it is a combination of the previous two models, where the candidates' points are selected uniformly at random, and for the voters we use the asymmetric Gaussian model. We use the same parameters for both of the distributions as above.

4. The *Asymmetric Gauss Candidate Positions, Uniform Voter Positions (GCP/UVP)* model, is a reversed version of the previous model, with the distributions swapped.

Radius Parameter

In this category we consider five ways of generating the radii for the candidates and the voters. We use the following models:

1. The *Gauss Candidate Radii, Gauss Voter Radii (GCR/GVR)* model, where we choose the radii from the Gaussian distribution with mean value 1.5 and standard deviation 0.5. We use the same distribution for the candidates and the voters.

2. The *Small Uniform Candidate Radii, Gauss Voter Radii (SUCR/GVR)* model. We pick the radii for the candidates uniformly at random from the interval $[0.7, 1.2]$, and for the voters we use the same Gaussian as in the previous model (with mean 1.5 and standard deviation 0.5).

3. The *Gauss Candidate Radii, Small Uniform Voter Radii (GCR/SUVR)* model. In this model we swap the distributions used in the previous one, that is, the radii for the candidates are picked from the Gaussian and the radii for the voters from the uniform distributions.

4. The *Small Uniform Candidate Radii, Small Uniform Voter Radii (SUCR/SUVR)* model. We pick the radii for both the candidates and the voters uniformly at random from the interval $[0.7, 1.2]$.

5. The *Large Uniform Candidate Radii, Large Uniform Voter Radii (LUCR/LUVR)* model. We pick the radii for both the candidates and the voters uniformly at random from the interval $[0, 3]$.

At first, it might be surprising why we do not include a constant radius as one of the distributions. We disregard this possibility because, then, all the profiles would be VR and CR (FCOP). Below in Proposition 2, we show why.

Proposition 2. *If every agent (candidate/voter) in a VCR profile has the same constant radius, then this profile satisfies both the VR and the CR properties (it is FCOP).*

Proof. Consider a VCR election $E = (C, V)$, where $C = \{c_1, ..., c_M\}$ and $V = \{v_1, ..., v_N\}$, and a constant radius R. An agent $a \in C \cup V$ is represented by a point x_a and a radius r_a, and in this special case every agent has the same radius $r_a = R$. We construct a CR election $E' = (C', V')$, where each voter $v' \in V'$ has radius $r_{v'} = 0$ and each candidate $c' \in C'$ has radius $r_{c'} = 2R$. We will show that these elections are identical. From the VCR definition, we know that for each candidate $c \in C$, if c is approved by a voter v_i, then

$$|x_c - x_{v_i}| \leq r_c + r_{v_i} \iff$$
$$|x_c - x_{v_i}| \leq R + R$$

Similarly, for each candidate $c' \in C'$ if c' is approved by a voter v'_i, then

$$|x_{c'} - x_{v'_i}| \leq 2R + 0$$

We showed that the elections E and E' are equivalent. Analogous reasoning works for the VR case, where each candidate has radius zero, completing the proof. □

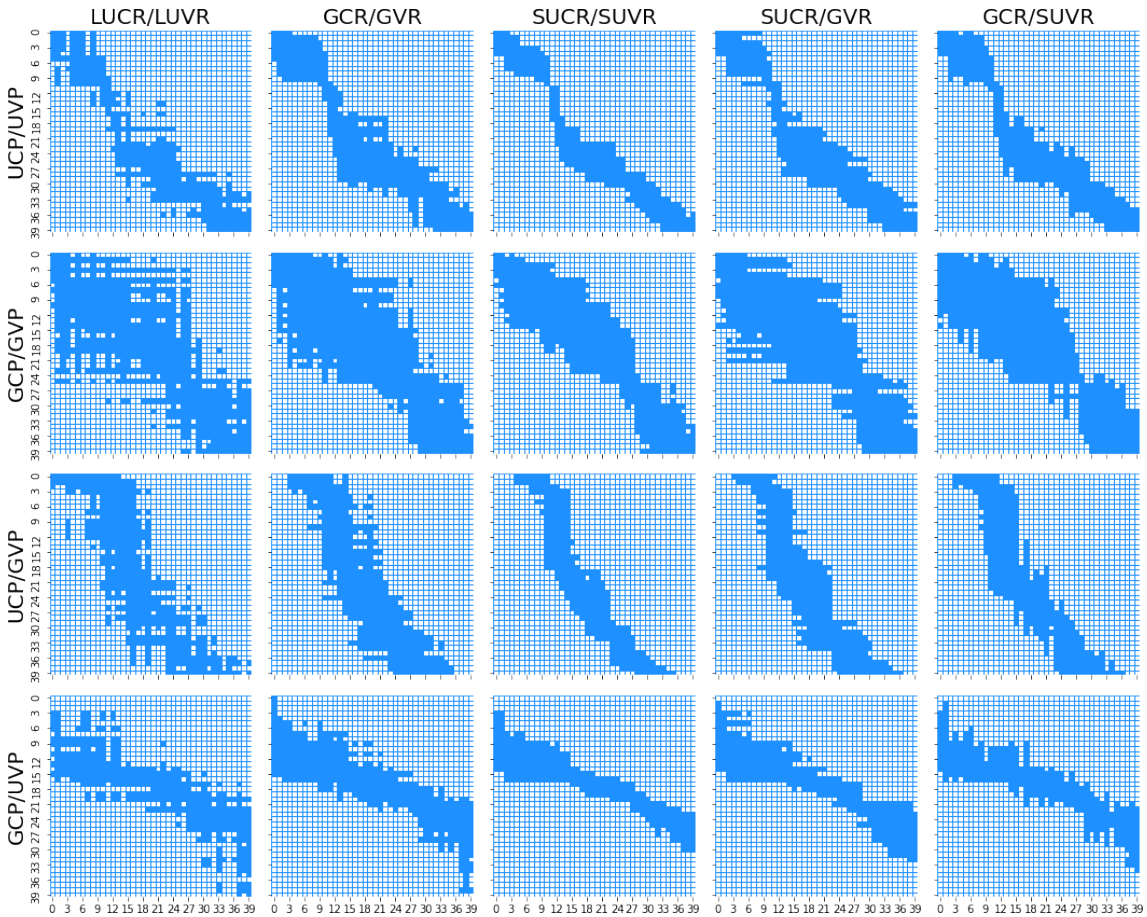


Figure 4.2 Examples of approval matrices for the elections of 40 candidates and 40 voters, from Experiment 2. Each subplot is a visual representation of an approval matrix of a selected profile. Solid color corresponds to 1 and white to 0 in cells of the matrix. To create a grid, we use radius candidate/voter models for the columns, and position candidate/voter models for the rows, e.g., top-left subplot is visualization of a profile generated with the LUCR/LUVR radius model and the UCP/UVP position model.

Results

First, in Figure 4.2, we present the visualization of our example profiles approval matrices and how their visual structure is impacted by the generation parameters. Each subplot represents a different combination of the generation parameters pairs. Columns correspond to the radius candidate/voter models, and rows to the position candidate/voter models.

Before, we further describe the results of the experiment, we need to observe trends in the visualization and explain some basic ideas. Based on the visualizations, we know that the VCR profiles produce semi-structured approval matrices. That is, the ones in an approval matrix create roughly continuous blocks (intervals), they form a cluster. However, in certain

circumstances, these blocks might have some irregularities (i.e., a 0 in sequence of 1's). The perfect example of this phenomenon are profiles from the first column in Figure 4.2, that is, profiles generated using the LUCR/LUVR radius model. We can spot the "holes" in otherwise regular structure of 1's. Another important trend in these visualizations is the general direction in which the block structure is skewed. For profiles in the first and second row of Figure 4.2 (i.e., the UCP/UVP and the GCP/GVP models), the skew is on a diagonal from the top-left to the bottom-right corner. However, profiles from the last two rows (i.e., the UCP/GVP and the GCP/UVP models), while maintaining general diagonal direction, they have a significant shift. The UCP/GVP ones are skewed vertically, opposite to the GCP/UVP ones, which are skewed horizontally.

In the following discussion, we tie these observations with the aggregated results of the experiment visualized in Figure 4.3 and Figure 4.4. In Figure 4.3 we show how many of the generated profiles have a given preference restriction. To understand the results better,

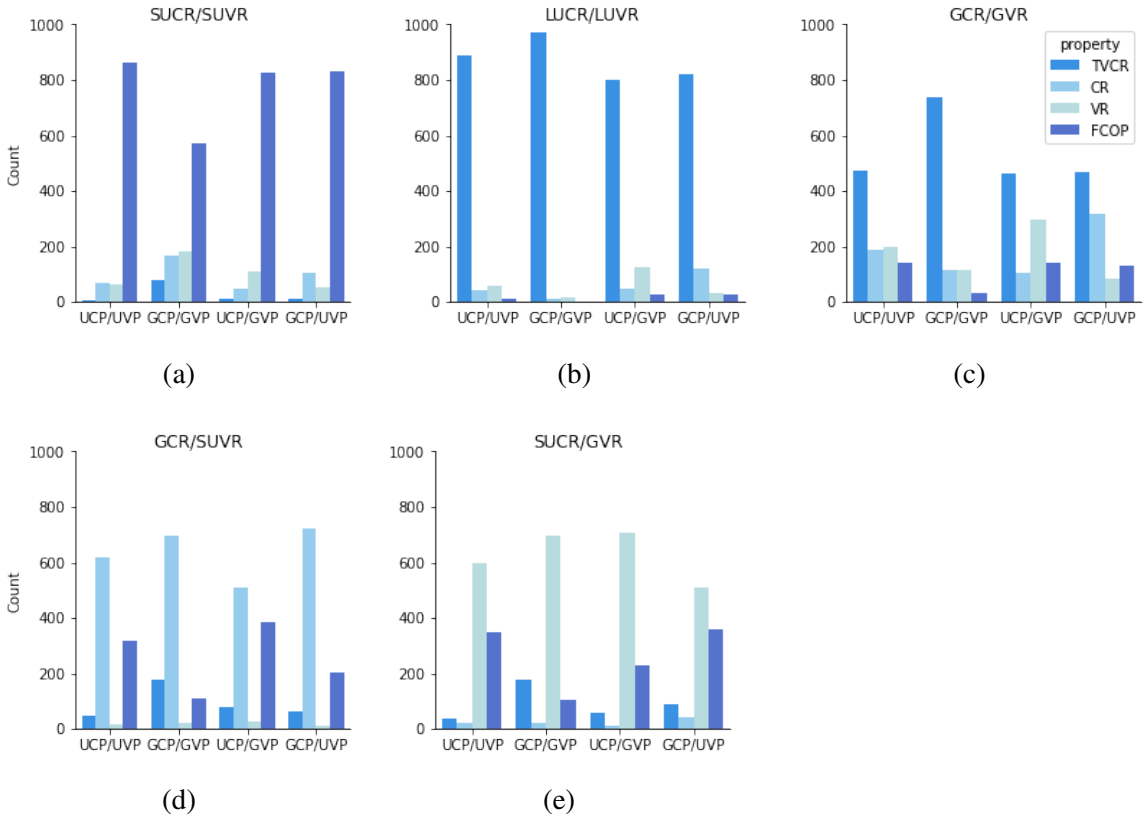
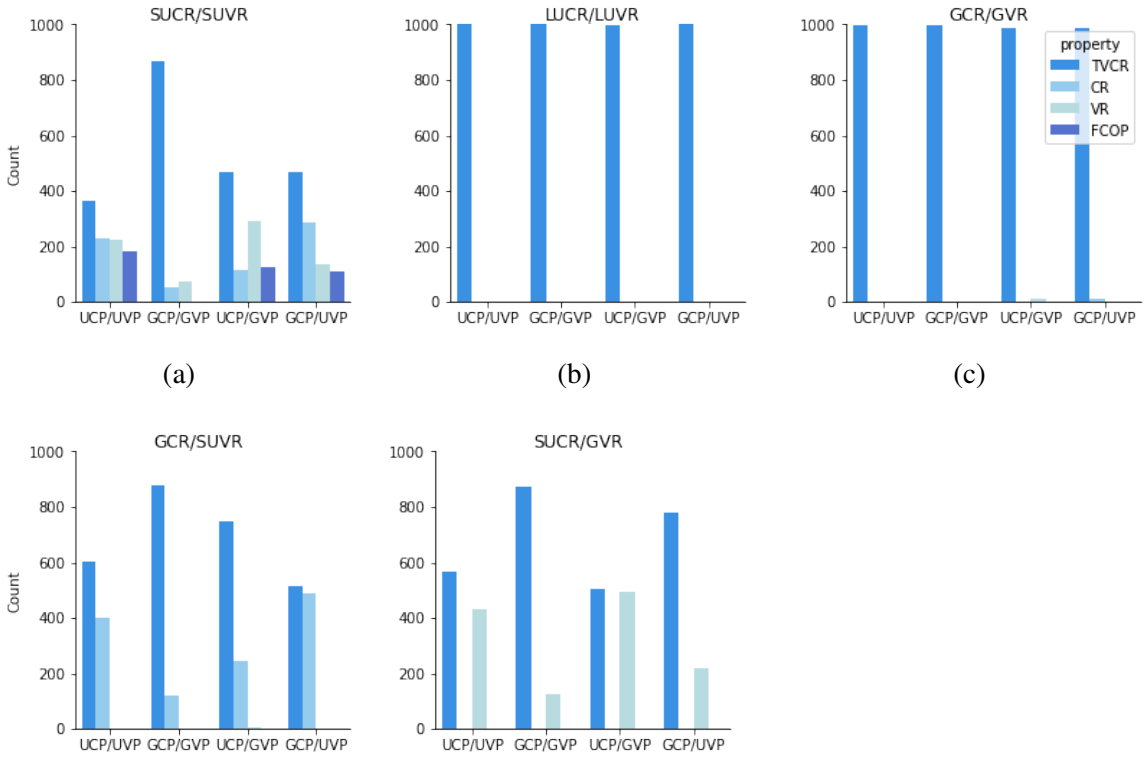


Figure 4.3 Number of profiles with a given property, for the elections of 20 candidates and 20 voters, from Experiment 2. Results are grouped by the parameters used for the profile generation. Each subplot corresponds to the selected radius candidate/voter model, and position candidate/voter models are represented on the x-axis.



(a) (b) (c)

(d) (e)

Figure 4.4 Number of profiles with a given property, for the elections of 40 candidates and 40 voters, from Experiment 2. Results are grouped by the parameters used for the profile generation. Each subplot corresponds to the selected radius candidate/voter model, and position candidate/voter models are represented on x-axis.

we split the collected data into groups. We use separate subplots to distinguish the radius candidate/voter models, and the position candidate/voter models are presented on the x-axis of each subplot. Figure 4.4 is almost identical, the only difference is the first one shows the results for the 20 candidates and 20 voters elections, whereas the second one describes the larger elections of 40 candidates and 40 voters.

Election Dimensions Our results for two election sizes have one crucial difference. Looking briefly at Figure 4.3 (20 candidates, 20 voters) and Figure 4.4 (40 candidates, 40 voters), we immediately see that in the larger elections the TVCR property is much more frequent. Election sizes in the experiment are relatively small when compared to real world scenarios. We can expect for this trend of increased number of TVCR profiles to continue for even larger elections. This result is very promising, as it strongly encourages further development and research related to the VCR domain.

In the subsequent discussion we will focus mostly on the results for smaller elections from Figure 4.3. We do so because there the differences between the parameters are more pronounced.

Position Choice. Let us analyze Figure 4.3c, which presents histograms for the GCR/GVR radius model and various candidate/voter position models. We consider this model to be our baseline scenario, because it has the most balanced distribution of number of profiles, between the selected preference restrictions. We compare the position models in pairs. Starting with the UCP/UVP and the GCP/GVP models, the latter one has noticeably more TVCR profiles. It can be easily explained by looking at the respective approval matrices in Figure 4.2 (2nd column, 1st and 2nd row). The profiles generated from the GCR/GVR model have higher count of approvals, which in turn increases the likelihood of any irregularities in the approvals block structure, which we directly link to the TVCR property.

Now, turning our attention to the second pair of the position models (still for the same radius model), the UCP/GVP and the GCP/UVP, we see that these are two perfectly symmetrical models. We only explain the first one, and keep in mind that the second one is simply a reversed scenario. Looking at the approval matrices in Figure 4.2 (3rd row), we notice that the blocks of approvals are skewed vertically, which is associated with the VR property, i.e., the approvals form intervals in each row of the matrix. On the contrary, in the second model (Figure 4.2 (4th row)) the blocks are skewed horizontally, which links to the CR property.

Radius Choice. From Figure 4.3, it is clear that the choice of the radius candidate/voter model can significantly reinforce or subside trends in the distribution of numbers of profiles with a selected property. Figure 4.3c was our baseline scenario when analyzing influence of the position candidate/voter models. Now, let us compare this subplot with the other ones. For example, the mixed radius models, that is, the GCR/SUVR (Figure 4.3d) and the SUCR/GVR (Figure 4.3e) models. Looking at the histograms for these models, we observe significant increase in the number of profiles with the CR and the VR properties, for the GCR/SUVR and the SUCR/GVR respectively. The core idea explaining this phenomenon is exactly the same as the one that we already described, when explaining the UCP/GVP and the GCP/UVP position models. Looking at approval matrices in Figure 4.2 (4th and 5th columns), the GCR/SUVR and the SUCR/GVR models output more consistent approval blocks (without 0's between the 1's), resulting in a large decrease of the number of the TVCR profiles. At the same time, these models work in synergy with the UCP/GVP and the GCP/UVP position models, and the created approval structures are skewed either horizontally or vertically, increasing the number of the VR or the CR profiles.

The striking difference between histograms for the LUCR/LUVR (Figure 4.3b) and the SUCR/SUVR (Figure 4.3a) radius models is again related to the inconsistencies in the approval blocks. They are easily observed in approval matrices visualized in Figure 4.2 (1st and 3rd column). Smaller radius range (the SUCR/SUVR) produces much more structured profiles, hence large number of the FCOP profiles. In contrast, large radius range (the LUCR/LUVR) outputs irregular approval blocks in the profiles, increasing the number of the TVCR profiles.

Summary. Most of the conclusions we draw are in line with the basic intuition. The results are very promising, we showed that the VCR domain can be significantly larger than the VR and the CR ones combined. We hope, this will encourage further development and research related to the VCR domain.

4.2 Transforming a TVCR Profile into a CR or a VR One

A natural question that arises from the previous experiments, is how different are the TVCR profiles from the VR or the CR ones? In other words, to what extent do we need to change a TVCR profile for it to become a VR or a CR one. To answer this question, we design yet another experiment.

4.2.1 Experiment 3: How different is a TVCR profile?

In this experiment we partially reuse the strategy for profile generation from the previous one. We narrow down our experiment parameters to only two radius generation methods, that is, the GCR/GVR and the LUCR/LUVR models. Nonetheless, we keep the position models unchanged and explore all four of them. Obviously, we only pick the TVCR profiles. To be exact, we use 400 of them for each category pair (in total, 8 pairs). Due to the limited computational resources, we only use profiles consisting of 15 candidates and 15 voters. Having a TVCR profile as input, we search for a minimal number of agents that we have to remove from this profile, for it to become a VR or a CR one. To achieve more meaningful results we categorize experiment parameters, and describe them below.

Agent Deletion Parameter. We consider three methods to transform a TVCR profile into a VR or a CR one:

1. The *Candidate Deletion (CD)* method, where we search for a minimal number d, such that, after deleting d candidates from the profile it becomes a VR or a CR profile.

2. The *Voter Deletion (VD)* method, which is similar to the CD method, but this time we delete d voters instead.

3. The *Candidate and Voter Deletion (CVD)* method, where we combine the two previous methods, and we simultaneously delete d_c candidates and d_v voters.

Additionally, we put an upper bound on the number of removed agents. For the CD and the VD methods the limit is 4, that is, we only try to remove 4 agents at most. If we fail to find a solution, we use a number 5 as a result, which we interpret as 5 or more agent deletions required to succesfully transform a profile. For the CVD method, the limit is 4 per each agent type, i.e, the maximum number of deleted agents is 4 candidates and 4 voters.

Transformation Result Parameter. Apart from choosing what type of agent (candidate / voter) to delete, for each profile we collect the transformation results in three groups. By the transformation result we mean the lowest possible cost of the transformation, i.e., the minimal number of agent deletions for a TVCR profile to become a VR or a CR one. We collect the following information:

1. The *Candidate Range (CR)*, where the cost is a number of deletions required to get a CR profile.

2. The *Voter Range (VR)*, where the cost is a number of deletions required to get a VR profile.

3. The *Candidate Range or Voter Range (CRVR)*, where the cost is a minimum from the previous two groups.

Results

We divide the observations in two groups based on the agent deletion parameter. First, we analyze the results for the CD and VD methods, and later we discuss the CVD method by itself. We use the similar ideas and concepts as in previous experiments to explain the results.

We present the results for the first group in Figure 4.5. The figure is split into six subplots. Each one of them is a visualization for different subset of results, divided by two parameters. First category is the radius candidate/voter model used to generate the profiles, and second one is based on the transformation result parameter, i.e, the CR, VR and CRVR types. A single subplot shows the distribution of a required number of agents to remove (a violin plot), to transform a TVCR profile into one with a given property.

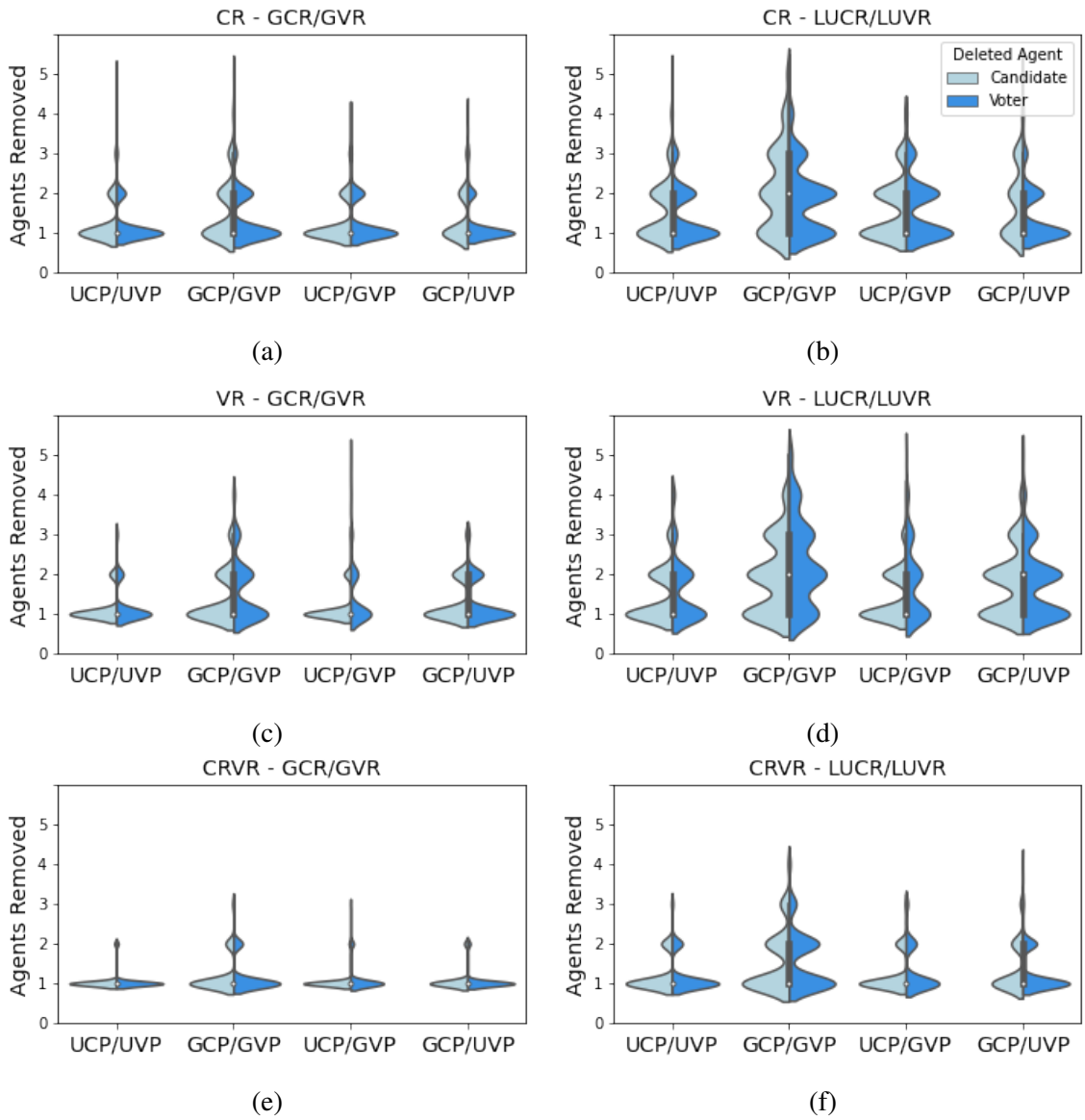


Figure 4.5 Required number of agents to remove, to transform a TVCR profile into a CR or a VR one, from Experiment 3. Each subplot consists of 4 violins, which are similar to a box and whisker plot. A violin plot shows the distribution of quantitative data, and uses kernel density estimation for the visualization. In our case, violins are additionally split in half, i.e., the left part (light color) shows a number of candidates deleted, and the right part (dark color) a number of voters deleted. Results are grouped by the experiment parameters. To create a subplot gird, we use radius candidate/voter models for the columns, and transformation result parameters for the rows, e.g., top-left subplot is a visualization of results for the GCR/GVR radius model and the CR property. Furthermore, position candidate/voter models are represented on x-axis.

Continuing, let us analyze Figure 4.5a and Figure 4.5c. They both show results for the GCR/GVR radius model, the first one corresponds to the CR and the second one to the VR properties. We notice that they are somewhat symmetrical. For example, for the GCP/GVP position model, to obtain the CR profile it is preferable to delete voters. However, if we look at the VR case, it is better to delete candidates. When saying preferred or better, we mean, it requires fewer agent removals. Another interesting observation in these plots, are the violins for the UCP/GVP and the GCP/UVP models. They are virtually swapped between the CR and the VR cases. To explain this, we go back to the respective histograms from Experiment 2 in Figure 4.3c. For the UCP/GVP model, the majority of profiles are VR (not counting the TVCR ones). This suggests that the TVCR profiles from this model are more similar to the VR ones than to the CR ones. This idea is reflected in the corresponding violin plots (Figure 4.5c), it is preferred to delete candidates to obtain the VR property, for profiles from the UCP/GVP position model and the GCR/GVR radius model. For the GCP/UVP position model we see the reversed scenario, in the histograms in Figure 4.3c the majority of profiles are CR, and the violin plots (Figure 4.5a) show that it is better to delete voters to obtain the CR profiles. We observe the same phenomenon for the LUCR/LUVR radius model in Figure 4.5b (the CR) and in Figure 4.5d (the VR). There is nothing unexpected in results for the CRVR property (Figure 4.5e and Figure 4.5f). Having said that, they give us a promising conclusion, that if we are indifferent whether we obtain the CR or the VR profile, we can greatly reduce the required number of agent deletions.

Let us discuss the second part of the experiment, that is results for the CVD method, in which we delete both the candidates and the voters simultaneously. We create plots in Figure 4.6 in the same way as in already described Figure 4.5. Only difference is conceptual, previously violins were split to distinguish between removed agent type, i.e, left half (light color) represented candidates and right half (dark color) showed voters. Now, the halves still represent the number of deleted agents, but process of deletion is simultaneous for candidates and voters. We note that if all the values in distribution are identical, then the violin plot is replaced with a single horizontal line. In most cases, it is enough to delete just one candidate and one voter to obtain a CR or a VR profile. Besides that, we explain why it appears that we need to delete more voters than candidates. It comes merely from implementation nuances. To find a number of required removals, we iterate thorough pairs of all (d_c, d_v) element combinations. That is we start by deleting one candidate and one voter, if there is no solution, then we try to delete one candidate and two voters, and only then if there is still no solution, we try two candidates and one voter, followed by 2 candidates and 2 voters, and so on.

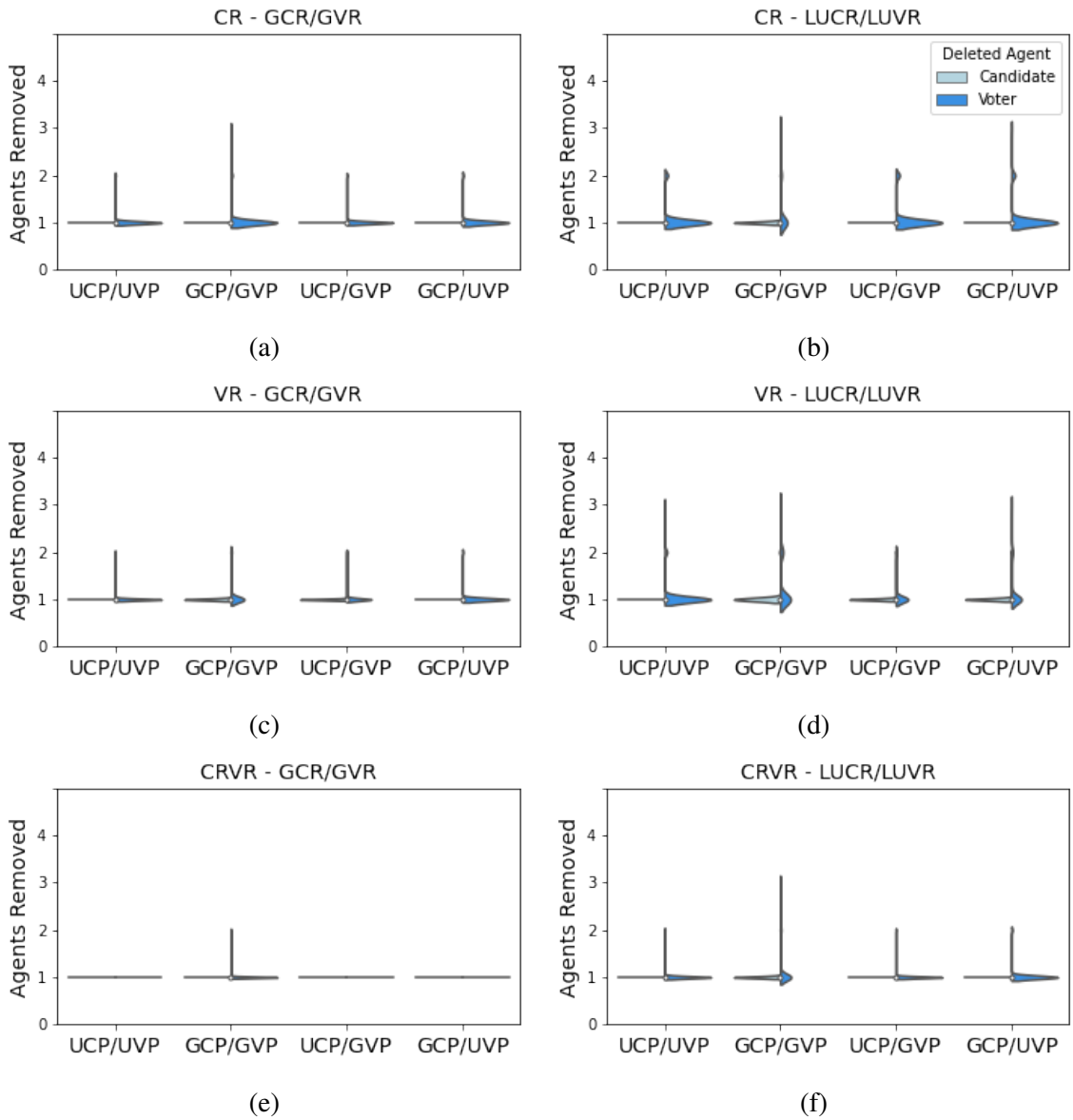


Figure 4.6 Required number of candidates and voters to remove, to transform a TVCR profile into a CR or a VR one, from Experiment 3. The plot is structured in the same way as in Figure 4.5, except one difference. Split in half violins no longer represent, independent deletion of the candidates or the voters, but correspond to simultaneous deletion of both, the candidates and the voters. If there is no violin (horizontal line), it means that all the values were identical.

In addition to the visual representation of the results we provide statistical metrics of the results, i.e., mean, standard deviation, minimum and maximum. Values are grouped by experiment parameters, in the same way as in the graphs, except the data is aggregated for all

the position models. We present them in Table 4.1 for the first part of the experiment, and in Table 4.2 for the second one.

Radius Model	Property	Deleted Agent	mean	std	min	max
GCR/GVR	CR	Candidate	1.366	0.633	1.000	5.000
		Voter	1.258	0.506	1.000	4.000
	CRVR	Candidate	1.089	0.304	1.000	3.000
		Voter	1.092	0.297	1.000	3.000
	VR	Candidate	1.256	0.513	1.000	4.000
		Voter	1.373	0.623	1.000	5.000
LUCR/LUVR	CR	Candidate	1.788	0.903	1.000	5.000
		Voter	1.587	0.737	1.000	5.000
	CRVR	Candidate	1.341	0.593	1.000	4.000
		Voter	1.336	0.567	1.000	4.000
	VR	Candidate	1.656	0.792	1.000	5.000
		Voter	1.851	0.935	1.000	5.000

Table 4.1 Statistical metrics for number of deleted agents, from Experiment 3.1

Radius Model	Property	Deleted Agent	mean	std	min	max
GCR/GVR	CR	Candidate	1.000	0.000	1.000	1.000
		Voter	1.021	0.146	1.000	3.000
	CRVR	Candidate	1.000	0.000	1.000	1.000
		Voter	1.001	0.025	1.000	2.000
	VR	Candidate	1.002	0.043	1.000	2.000
		Voter	1.022	0.148	1.000	2.000
LUCR/LUVR	CR	Candidate	1.002	0.043	1.000	2.000
		Voter	1.107	0.332	1.000	3.000
	CRVR	Candidate	1.001	0.035	1.000	2.000
		Voter	1.029	0.171	1.000	3.000
	VR	Candidate	1.008	0.090	1.000	2.000
		Voter	1.127	0.357	1.000	3.000

Table 4.2 Statistical metrics for number of deleted agents, from Experiment 3.2

4.3 WIP: Summary

First, we showed how results from Experiment 3 further confirm our conclusions from Experiment 2. Moreover, our results provide strong basis, to consider transforming a TVCR profile into a CR or a VR one, if it would be required by other applications.

Chapter 5

Control Problem

In this chapter we demonstrate how a certain control problem that is known to be NP-hard for unrestricted preferences, becomes polynomial-time solvable for the VCR elections. First in Section 5.1, we discuss variants of the control problem. Additionally, we present two existing polynomial-time algorithms for a control problem, for elections in the VR and the CR domain. Later in Section 5.2, we describe our solution for the VCR elections, and we show that it unifies the two aforementioned algorithms for the VR and the CR cases.

5.1 Related Work (WIP: Inny tytuł?)

Control problems were first introduced by Bartholdi et al. (1992), and later extended by Hemaspaandra et al. (2009). The control refers to an act of manipulating the structure of an election in a certain way. We assume that there is some person or authority who is in charge of organizing the election. We refer to them as the *election chair*. The chair has a goal to change the outcome of the election. Formally we call it the *control type*, and we distinguish two types. The first one, the *constructive* control (CC), where the chair's goal is to ensure a victory of a selected candidate. The second one, the *destructive* control (DC), which is an opposite, that is, the chair's intention is to prevent a selected candidate from winning. Furthermore, control problems are grouped by the method used to change the structure of an election, we call it the *control action*. Usually in literature, for each control type (i.e, the CC or the DC) we consider four possible actions, that is, adding candidates (AC), deleting candidates (DC), adding voters (AV) or deleting voters (DV). For further reading, see the chapter by Faliszewski et al. (2016), which provides a complete overview of the control problem and the closely related bribery problem as well.

The algorithms that we discuss in the following sections, consider the constructive control by deleting voters (CC-DV), and of course, for the winner determination, we use the approval voting rule (see Definition 2.1.1). Below, we give a formal definition of our problem:

Definition 5.1.1. *We are given a set of candidates C, a set of voters V, a nonnegative integer k, and a selected candidate $p \in C$. In the constructive control by deleting voters problem for the approval voting (Approval-CC-DV), we ask whether there exists a set $V' \subseteq V$, such that p is the unique winner of the approval voting election $E = (C, V \setminus V')$, where $|V'| \leq k$.*

Now, we define some basic concepts and terminology, that will help us to explain the algorithms in the subsequent sections. First, we introduce an alternative representation of an agent in a VCR profile.

Definition 5.1.2. *In the VCR model, every agent a is represented by their point x_a and radius r_a. We define a's range beginning and range end as:*

$$b(a) = x_a - r_a,$$

$$e(a) = x_a + r_a.$$

Additionally, to represent the AV-score difference between two candidates, we use:

Definition 5.1.3. *Having an approval election E and two candidates a and b. Under the approval-voting scoring rule, we say that:*

$$scoreDelta(E, a, b) = score_{AV}(E, a) - score_{AV}(E, b).$$

However, we note that we use a more concise notation, that is, the $scoreDelta(a, b)$.

5.1.1 Constructive Control by Deleting Voters For the VR Elections

This algorithm is presented in the work of Faliszewski et al. (2009). It is the first paper to consider solving control problems under preference restrictions. The restriction that they consider is the single-peaked one. However, in the approval elections it is virtually identical to the voter range model (i.e., candidates' radii are zero). We only present a brief description of the algorithm, that explains the core idea and some necessary concepts. For the detailed explanation, proofs and examples we point to the source work.

The algorithm deploys a greedy strategy. In the explanation we follow the example from Figure 5.1. Let us consider a problem of the CC-DV under the approval voting rule. Having a set of candidates C in the VR order, a set of voters V, a designated winner candidate p, and

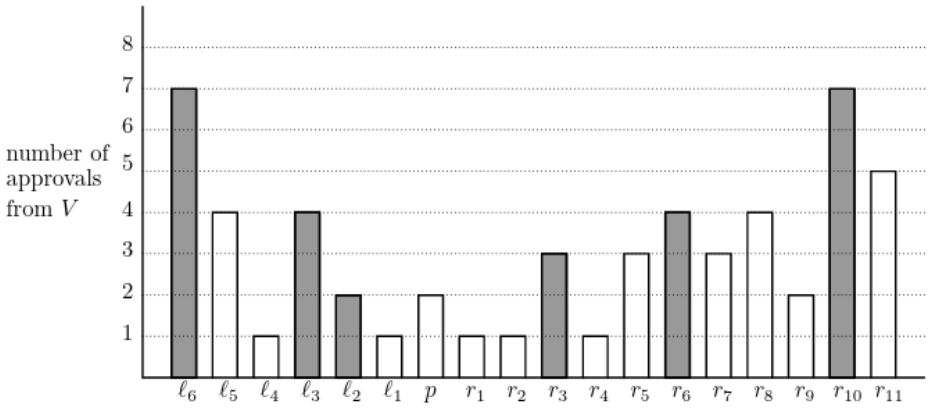


Figure 5.1 Illustration of the CC-DV for the VR elections from Faliszewski et al. (2009). Bar chart of approvals from V. The candidates are in the VR order, and each candidate is labeled with respect to whether he or she is left or right of the candidate p. Additionally, the "dangerous" candidates are marked with shaded bars.

a maximum number of voter deletions k. To find a solution (if it exists) we take the following steps: First, we group the candidates based on their position relative to p, i.e., left or right from p. Then, we proceeded to selecting the dangerous candidates (opponents). The first opponent to the right of p is the leftmost candidate from the right group, that gets at lest as many approvals as p (i.e., r_3). Every next right dangerous candidate is the candidate that gets strictly more approvals than the previous dangerous candidate (i.e., r_6, and then r_{10}). We take analogous steps for the candidates on the left. From this point, we try to defeat dangerous candidates. We process one at a time in each iteration. We start with the closest right opponent (i.e., r_3), and call him c_R. Now we have to choose which and how many voters to delete. We consider only a subset of voters, such that each voters' range beginning is in the open-closed interval between p and c_R, i.e., $\widehat{V} = \{v \in V \mid b(v) \in (p, c_R]\}$. Next we try to delete $scoreDelta(c_R, p) + 1$ rightmost voters from $\widehat{V}$, that is the ones with the highest range end (i.e., $e(v)$). If we fail, or we exceed the voter deletion limit k, the solution does not exist. We repeat these steps until the candidate p becomes a unique winner (or we fail in the process). We recalculate the approvals and the dangerous candidates set after each iteration. Once we reached the point where all the dangerous candidates are in the left group, then we simply "flip the universe", and consider them to be the right ones.

5.1.2 Constructive Control by Deleting Voters For the CR Elections

The next algorithm is due to Magiera and Faliszewski (2014). They study multiple election types, but we discuss only the approval elections case. They consider control problems for the single-crossing preferences restriction. Which, in the approval elections, is virtually

identical to the candidate range model (i.e., voters' radii are zero). We again provide only a brief explanation of the algorithm, and for specifics we refer to the original paper. This algorithm has many similarities to the previous one, mainly it utilizes a greedy strategy as well. Let us consider a problem of the CC-DV under the approval voting rule. Having a set of candidates C, a set of voters V in the CR order, a designated winner candidate p, and a maximum number of voter deletions k. For each candidate $c \in C$, we say that the voters who approve of c form an interval: $v_{s(c)}, v_{s(c)+1}, ..., v_{f(c)}$ (the CR property). Additionally, we define a set of dangerous candidates: $\widetilde{C} = \{c \in C \setminus \{p\} | \, scoreDelta(c,p) \geq 0\}$, and a subset of voters who do not approve of p: $V' = \{v \in V | \, p \notin A(v)\}$. The rest of the algorithm is straightforward. It executes the following operation until p is a winner, or we fail by reaching the limit of deletions k. First, we need to select a dangerous candidate that we will try to defeat. To do so, we find an opponent $c_N \in \widetilde{C}$, for whom $f(c_N)$ is the smallest. Then, we try to delete $scoreDelta(c_N, p) + 1$ voters, starting with the voter at the end of interval (i.e., $v_{f(c_N)}$). If we fail, the solution does not exist. After each iteration of this step, we recalculate the sets $\widetilde{C}$ and V'.

5.2 Constructive Control by Deleting Voters For the VCR Elections

This time, we describe our algorithm in full, and consider all the caveats. For the sake of completeness, we will define some notions once again.

Let us consider a problem of CC-DV under the approval voting rule. Having a VCR election, with a set of candidates C, a set of voters V, a designated winner candidate p, and a maximum number of voter deletions k. First, we discuss some concepts and observations, that will help us explain the algorithm afterwards. We use an alternative representation of an agent in the VCR election, that is, by his or hers range beginning and end (see Definition 5.1.2). Next, for p to become a unique winner, we need to defeat every candidate that has at least as many votes as p. We call this subset of candidates a set of dangerous candidates (opponents), formally:

$$\widetilde{C} = \{c \in C \setminus \{p\} | \, scoreDelta(c,p) \geq 0\}.$$

Another observation is that we never delete a voter who approves of p, clearly it would not bring us any closer to a solution. When deleting voters, we consider only the ones from a set:

$$V' = \{v \in V | \, p \notin A(v)\}.$$

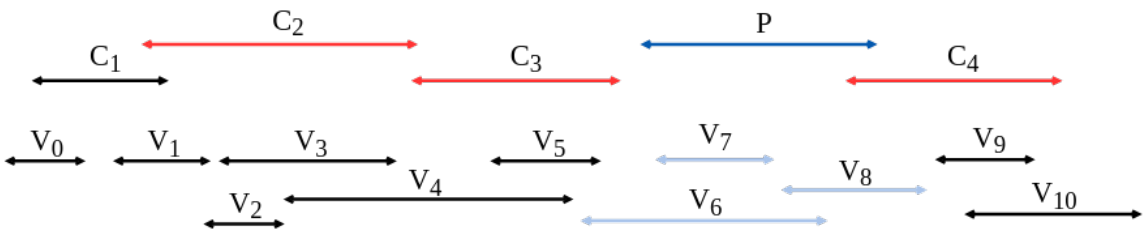


Figure 5.2 A VCR election for the CC-DV problem. The candidates are labeled with C_i (expect for the preferred candidate P), and the voters with V_i. We use colors to distinguish some key groups of agents. That is, P is in dark-blue, and voters that approve of him or her (we never delete them) are in light-blue color. Additionally, all the initially dangerous candidates are in red.

Our algorithm is a greedy one. It repeats the following steps until p becomes a unique winner, or it fails by exceeding the maximum number of voter deletions k. First, we find a dangerous candidate that we will try to defeat. Then, to achieve this goal, we decide which voters to delete. After each iteration, we recompute approvals and the set of dangerous candidates $\widetilde{C}$. We describe these steps in detail below:

1. **Choosing opponents.** The only candidates that we consider are from the set of opponents $\widetilde{C}$. We pick the leftmost candidate, that is, the one with the minimal beginning of the range $b(c)$ (we break ties arbitrarily). We refer to the selected candidate as c_L.

2. **Deleting voters.** Having currently processed opponent c_L. We consider only the voters who approve of him or her, i.e., $\widehat{V} = \{v \in V' | \, c_L \in A(v)\}$. To defeat c_L, we need to delete $scoreDelta(c_L, p) + 1$ voters from $\widehat{V}$. If $|\widehat{V}| < scoreDelta(c_L, p) + 1$, then we fail (it is impossible to delete enough voters to defeat c_L). We proceed with deleting the rightmost voters, that is, the ones with the maximum end of the range $e(v)$ (we break ties arbitrarily). By choosing the rightmost voters, we try to maximize the number of other dangerous candidates who might be contained inside each voter's range. We can do so because of our leftmost candidate picking strategy. At this point, any dangerous candidate to the left of c_L would have already been defeated.

We illustrate the algorithm in Example 5, which describes an election from Figure 5.2.

Example 5. *Let us discuss an instance of the CC-DV problem. Having the VCR election from Figure 5.2, the designated candidate P with 3 votes (i.e., $\{V_6, V_7, V_8\}$) and the set of dangerous candidates $\widetilde{C} = \{C_2, C_3, C_4\}$. We proceed to find the smallest number of voters that we need to delete for P to win the election. First, we choose the leftmost dangerous candidate C_2. We need to delete two voters ($scoreDelta(C_2, P) + 1$). We pick the rightmost*

voters V_4 and V_3. Now, an important observation is that by removing V_4 we defeated next dangerous candidate C_3. We finish this iteration, and recompute approvals and the set $\widetilde{C}$. The only dangerous candidate left is C_4, and we just need to delete 1 voter. We pick V_{10} (in this case it would not matter if we chose V_9). We succeed, P is the winner, and in total we removed three voters, i.e, V_4, V_3 and V_{10}.

In the subsequent sections, first, we discuss a certain special case for which our voter picking strategy might not be obvious. Later, we provide informal sketch proofs that show how our algorithm is equivalent to the VR and the CR ones. Finally, we give a formal proof of correctness of the algorithm.

5.2.1 Special Case of Contained Candidates

In the VCR model it is possible that some candidates have their interval fully contained inside other candidate's interval. If so, every voter who would approve the smaller candidate (i.e., internal) would also approve the larger one (i.e., external), however not vice versa. For this case, our algorithm's strategy of choosing rightmost voters to delete, might seem suboptimal at first glance. We define additional terminology and symbols to help us explain why we do not have to worry about this special case.

Definition 5.2.1. *Having an instance of the CC-DV problem, with an approval election E, a set of candidates C, and the designated winner candidate p. We extend representation of the score difference between two candidates (see Definition 5.1.3). For each candidate $c \in C \setminus \{p\}$ we represent a number of voter deletions required for p to defeat c with:*

$$\Delta(c) = score_{AV}(E,c) - score_{AV}(E,p) + 1.$$

Definition 5.2.2. *Having an election E, where a candidate is contained inside another one. The two candidates are, the one with larger range c_E, and the one with smaller range c_I. The voters who approve of c_E or c_I are in a set $\widehat{V}$. We divide the voters into two groups. First, a subset of voters who approve of c_E but not c_I:*

$$\widehat{V}_E = \{v \in \widehat{V} \mid c_E \in A(v) \wedge c_I \notin A(V)\},$$

and the remaining ones:

$$\widehat{V}_I = \widehat{V} \setminus \widehat{V}_E.$$

We know the size of these subsets based on the approval scores of each candidate:

$$|\widehat{V}_E| = score_{AV}(E,c_E) - score_{AV}(E,c_I),$$

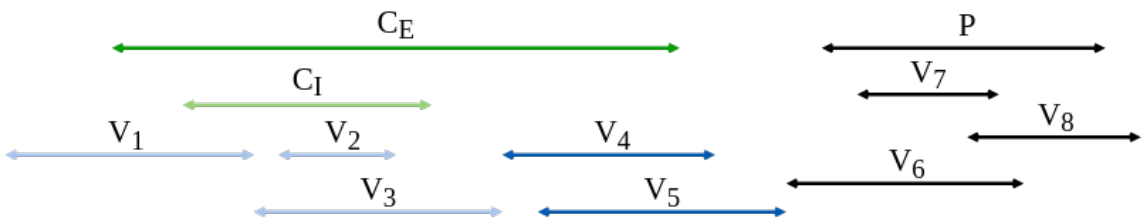


Figure 5.3 A VCR election with two candidates contained inside each other's ranges. The larger one c_E (external), and the smaller one c_I (internal). Furthermore, voters from the set $\widehat{V}$, are divided into two groups, i.e., $\widehat{V}_E = \{V_4, V_5\}$ a subset of voters who approve of c_E but not c_I, and a subset of remaining voters $\widehat{V}_I = \{V_1, V_2, V_3\}$. Because we discuss the CC-DV problem, there is also the preferred candidate p and his voters.

$$|\widehat{V}_I| = |\widehat{V}| - |\widehat{V}_E|.$$

Let us consider an instance of the CC-DV problem, with an election that has the special case. We know that a contained candidate c_I has as much or less total votes than the larger candidate c_E (i.e., $|\widehat{V}| \geq |\widehat{V}_I|$). The internal candidate c_I might not even be a dangerous candidate at all (i.e., $\Delta(c_I) \leq 0$). For p to defeat c_E we need to delete $\Delta(c_E)$ voters, and to defeat c_I (independently of c_E), we need to delete $\Delta(c_I)$ voters (see Definition 5.2.1). Now we show that in the process of defeating the larger candidate we always automatically defeat the smaller one.

$$\begin{aligned} \Delta(c_E) = \\ score_{AV}(c_E) - score_{AV}(p) + 1 = \\ |\widehat{V}_E| + score_{AV}(c_I) - score_{AV}(p) + 1 = \\ |\widehat{V}_E| + \Delta(c_I). \end{aligned} \tag{5.1}$$

Using simple transformations in Equation 5.1 based on Definition 5.2.2, we show that:

$$\Delta(c_E) = |\widehat{V}_E| + \Delta(c_I).$$

It means that to defeat c_E we have not other choice but to delete at least $\Delta(c_I)$ voters from $\widehat{V}_I$, the remaining $|\widehat{V}_E|$ voters can be deleted from any of the subsets (i.e., $\widehat{V}_E$ or $\widehat{V}_I$). This will result in defeating both c_E and c_I (automatically).

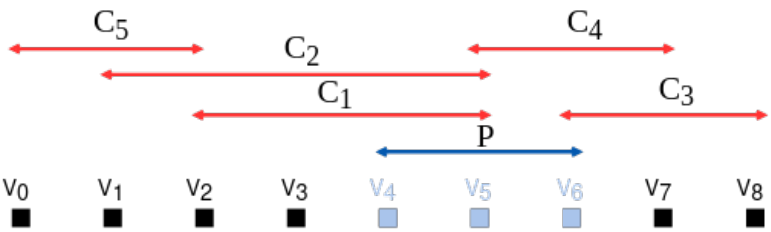


Figure 5.4 A CR election for the CC-DV problem, based on an example from Magiera and Faliszewski (2014). The candidates are labeled with C_i (expect for the preferred candidate P), and the voters with V_i. We use colors to distinguish some key groups of agents. That is, P is in dark-blue, and voters that approve of him or her (we never delete them) are in light-blue color. Additionally, all the initially dangerous candidates are in red.

5.2.2 Sketch Proof For the CR Elections

We show how our CC-DV algorithm for the VCR elections is equivalent to the one from Magiera and Faliszewski (2014) for the CR elections. Let us consider an instance of the CC-DV problem for a CR election (agents are still represented using the VCR model). In the explanation we follow the example from Figure 5.4.

Choosing opponents

In the CR algorithm, we choose dangerous candidates to process ordered by their voter's interval end (i.e., $v_{e(c)}$) ascending. We can see that it is equivalent to picking the leftmost candidate (i.e., min $b(c)$) from the set of opponents $\widehat{C}$ in the VCR algorithm. The selected candidate will also have the smallest $v_{e(c)}$ (candidate C_5 in the example). Of course the above statement is not true for the special case of the contained candidates. However, in Section 5.2.1, we already explained why it is irrelevant for the overall result.

Voter deletion

Now, once we now how to pick an opponent c_L, we show that choosing voters is identical as well. Notion of the beginning/end of the voter's interval in the CR model is equivalent to the left/right end of voter's range in the VCR setting. Both algorithms try to delete $scoreDelta(c_L, p) + 1$ voters, the CR one starts deleting voters from the end of interval, while the VCR one does virtually the same by deleting the rightmost voters first (i.e., max $e(v)$).

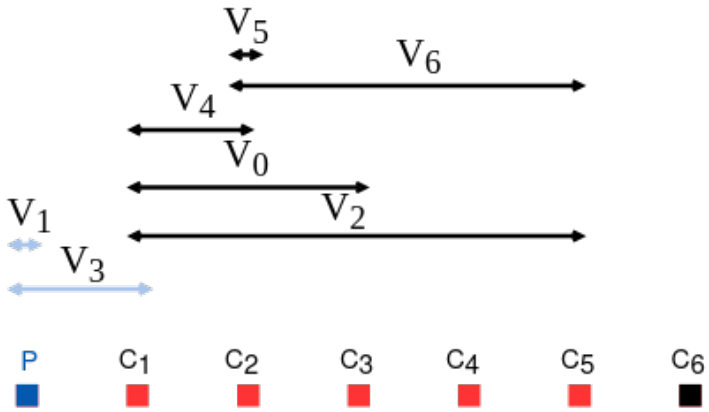


Figure 5.5 A VR election for the CC-DV problem. The candidates are labeled with C_i (expect for the preferred candidate P), and the voters with V_i. We use colors to distinguish some key groups of agents. That is, P is in dark-blue, and voters that approve of him or her (we never delete them) are in light-blue color. Additionally, all the initially dangerous candidates are in red.

5.2.3 Sketch Proof For the VR Elections

We show how our CC-DV algorithm for the VCR elections is equivalent to the one from Faliszewski et al. (2009) for the VR elections. Let us consider an instance of the CC-DV problem for a VR election (agents are still represented using the VCR model). In the explanation we follow the example from Figure 5.5. Strategy of the VR algorithm is significantly different from the VCR one (when compared to the CR one). However, if we introduce one simple assumption and an additional step to our algorithm, they become identical. First, let us assume that the preferred candidate p is always the first candidate in the VR order (see election in Figure 5.5). This assumption is correct based on two facts:

1. Having candidates in order $[c_L, p, c_R]$, if there was a voter v who approved of c_L and c_R it would also have to approve of p (because of the VR property), so we could not delete v anyway.

2. Next, for the VCR algorithm (with the above assumption) to be complete, we need to extend it by using the "universe flip" method from the VR algorithm. That is, we process only the dangerous candidates to the right of candidate p, and once there is none left, we proceed with the remaining ones on the left side of p, treating them as if they were on the right side.

Choosing opponents

We can see that the VCR algorithm strategy (the extended one) of picking the leftmost dangerous candidates is identical to the VR algorithm choosing the first candidate on the right side of p. However, there is another difference between the algorithms, the VCR one considers every candidate with score equal or higher than p to be a dangerous candidate (i.e., $\{C_1, C_2, C_3, C_4, C_5\}$ in the example). Whereas the CR one limits this set, by picking only candidates that have strictly more votes than a previously picked dangerous candidate (i.e., $\{C_1, C_2\}$ in the example). Nonetheless, in the VCR algorithm we recompute the set of dangerous candidates after each iteration (we defeat $\{C_3, C_4, C_5\}$ as a side effect of defeating C_1), which ensures that we never process any unnecessary dangerous candidates.

Voter deletion

The reasoning here is very similar to the one in the voter deletion part of the CR sketch proof. Both algorithms try to delete $scoreDelta(c, p) + 1$ voters. The VR algorithm starts deleting voters from the end of interval (max $e(v)$). The VCR one does virtually the same by deleting the rightmost voter first.

5.2.4 Proof For the VCR Algorithm

Now, we provide a formal proof of correctness of our VCR algorithm. The proof requires additional definitions and notation that we introduce below.

Definition 5.2.3. *Having a solution S to the CC-DV problem, that is, a set of k deleted voters. We define the score of this solution as a sum of removed voters' range ends:*

$$score(S) = \sum_{i=1}^{k} e(v_i)$$

(WIP: Czy możemy zdefiniować jakiś alternatywny wynik, czy wynik musi być po prostu równy k?)

Proposition 3. *The VCR CC-DV algorithm finds an optimal solution in polynomial time.*

Proof. Let S be the solution found by the algorithm, and S^* an optimal solution to the problem. We will prove that $S = S^*$ by contradiction. Assume $S \neq S^*$, therefore, $S - S^* \neq \emptyset$. Let v, be any voter in $S - S^*$. When v was added to S, it was the rightmost voter ($\max(e(v))$). Since S^* is optimal, then there must be some voter $x \in S^*$, such that $e(x) < e(v)$, because v is the rightmost voter. Let $S^{*\prime} = S^* \cup \{v\} - \{x\}$, we know that $S^{*\prime}$ is a correct solution. (WIP:

Dlaczego $S^{*\prime}$ jest poprawny? - Czy to właśnie tu musimy uzasadnić specjalny przpadek "wewnętrznych kandydatów"? Czy może być to jakiś osobny, wcześniej pokazany lemat?)
However,

$$score(S^{*\prime}) = score(S^*) + e(v) - e(x) > score(S^*)$$

contradicting that S^* is an optimal solution. We have reached a contradiction, so our assumption must have been wrong. Thus $S = S^*$, hence S is an optimal solution. □

5.2.5 Constructive Control by Adding Voters For the VCR Elections

We give only a sketch of an algorithm for the constructive control by adding voters variant.

Definition 5.2.4. *We are given a set of candidates C, a set of voters V and a set of unused voters W, a nonnegative integer k, and a selected candidate $p \in C$. In the constructive control by adding voters problem for the approval voting (Approval-CC-AV), we ask whether there exists a set $\widetilde{V} \subseteq W$, such that p is the unique winner of the approval voting election $E = (C, V \cup \widetilde{V})$, where $|\widetilde{V}| \leq k$.*

Our strategy is based on the similar solution from Magiera and Faliszewski (2014). It comes down to adding all the unused voters from W to the election, and then solving a reversed problem of the constructive control by deleting voters.

1. Add all $v \in W$ to the election
2. Is p the winner?
3. Try to delete $k' = |W| - k$ voters from W.

Chapter 6

Summary

6.1 Conclusions

6.2 Future Work

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